Lab 3: Architectural Pattern Selection – Written Justification

Project: Patient Health Record Consent Management System

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Architecture Selection: "We chose Layered Architecture for the Patient Health Record Consent Management System."

Architectural Choice

We selected a **Layered Architecture**, organised into three layers as shown in the component diagram. The **Client Layer (Web & Mobile UI)** contains Patient UI, Admin Console, and Doctor Portal. The **Application Services Layer (Business Logic)** contains the Consent Management, Doctor Verification, Clinical Record Access, and Audit Trail Logging Components. The **Persistence Layer (Databases & Secure Storage)** contains the Consent & User DB, Append-Only Audit Store, and EHR/Diagnostic Records Repository. Layers communicate downward through named interfaces – ConsentService, VerifyDoctor, RecordAccess, AuditLogging, UserDataAccess, AppendOnlyLog, and EHRDataAccess.

Reason 1: Clear Separation of Concerns for a Well-Defined Workflow

The system's real usage follows a top-down path: a patient grants or revokes consent, an administrator verifies a doctor, and a doctor views records, with each action flowing from the Client Layer into the Application Services Layer before ever reaching storage. A Layered Architecture mirrors this sequence directly, making the system easy to understand, build, and reason about for a single, well-scoped application rather than a set of independently distributed services.

Reason 2: Enforces a Single, Predictable Path to Sensitive Data

The diagram's <<use>> dependencies make this explicit: the Doctor Verification Component checks verification with Consent Management, and Clinical Record Access validates consent with Doctor Verification, before any record is accessed. In a Layered Architecture, the Persistence Layer is only ever reached through the Application Services Layer above it, so there is no direct path from the Client Layer straight into the EHR/Diagnostic Records Repository or Consent & User DB.

Security Advantage

Because Clinical Record Access depends on Doctor Verification, and Doctor Verification in turn checks with Consent Management, access to the EHR/Diagnostic Records Repository (via EHRDataAccess) can only happen after both a verified doctor and a valid consent record have been confirmed. The Client Layer never talks to the Persistence Layer directly, so a flaw in Patient UI or Doctor Portal cannot bypass these checks to reach sensitive medical data.

Performance Benefit

Because the Audit Trail Logging Component sits alongside the other Application Services and is called directly for "log event" and "log access" actions, writes to the Append-Only Audit Store happen in the same request path without extra network hops. For a system of this size, this keeps everyday operations such as verifying a doctor or granting consent fast, without the added latency a more distributed architecture would introduce.

Screenshot of Component Diagram:-

